

SMART PERSONNEL NEWSPAPER

SOFTWARE DESIGN DOCUMENT

- 1) Övgü Bejan Taburoğlu – 230201020 (Project Manager)
- 2) Azra Efza Gür – 220201025
- 3) Elif Köken – 220201054
- 4) Fevzi Samed Ünal – 220205032
- 5) Hamit Güneş – 220204042
- 6) İbrahim Türkel – 220201047

1. Use Case Design

1.1. User-Goal Level Use Cases

- **UC-1: Manage Profile and Preferences**

Create: User creates initial profile/preferences during first setup.

Read: System displays current profile information and selected interests.

Update: User updates name, interests, language, notification or preference settings.

Delete: User clears/removes selected preferences or resets preferences to default.

- **UC-2: View Personalized Feed**

Read: User views personalized articles, source references, dates, trending news, and integrated events.

- **UC-3: Search and Filter News**

Create: User may create/save a search query or bookmark/save an article, if this feature is included under this use case.

Read: User searches and reads filtered news results by keyword, category, date, source, or popularity.

Update: User changes active filters or sorting options.

Delete: User clears filters, removes saved searches, or removes saved/bookmarked articles.

- **UC-4: Export and Print Hardcopy**

Read: User previews the selected newspaper edition before printing. Update: User changes layout options such as A4, tabloid, booklet, or selected articles before export.

Delete: User cancels export or temporary generated PDF files are removed by the system.

- **UC-5: View Institutional Events and Announcements**

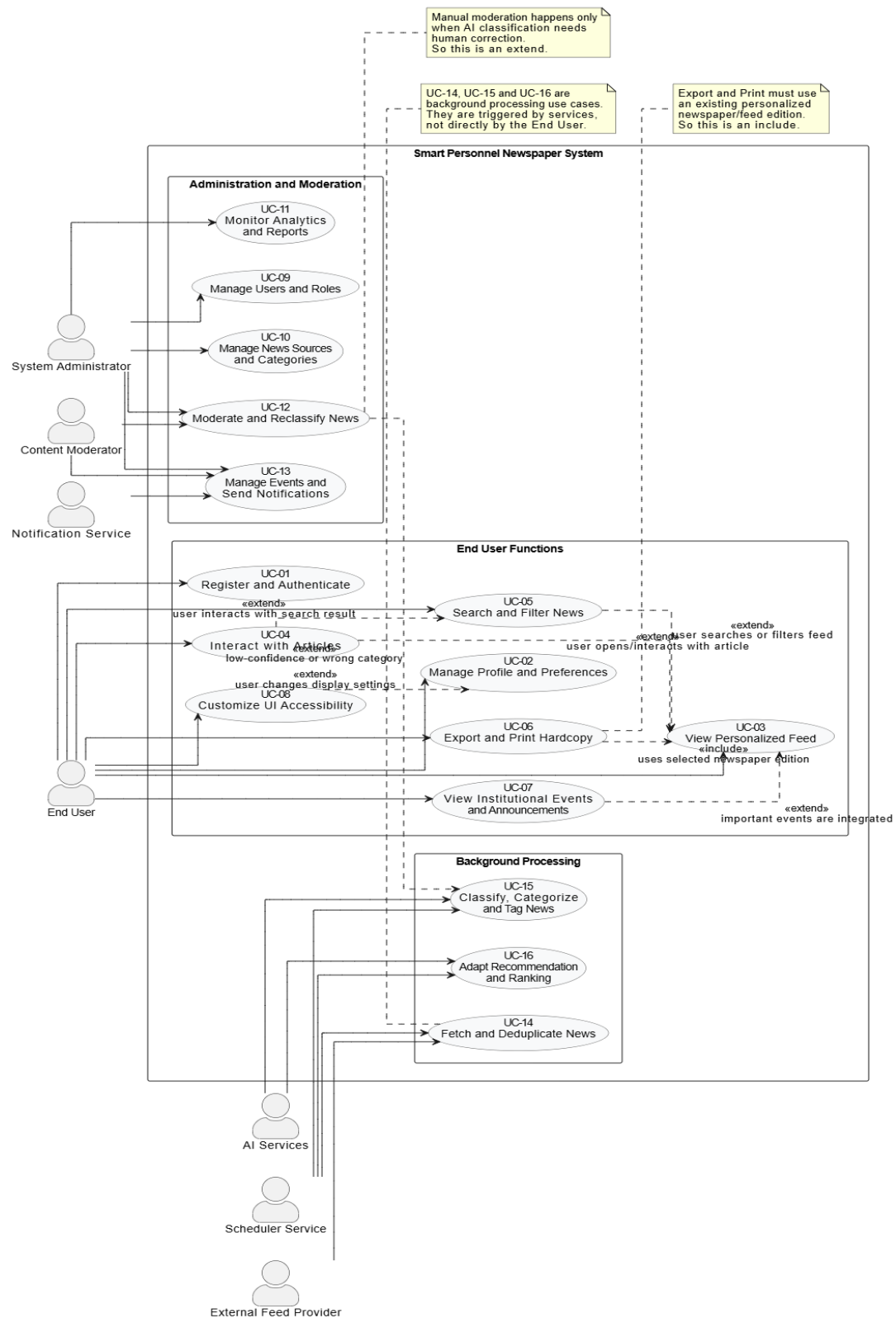
Create: User may create a reminder/notification preference for an event, if supported.

Read: User views institutional events, announcements, deadlines, exams, and event details.

Update: User may mark an announcement as read or update reminder status.

Delete: User may dismiss/hide a notification, but actual event deletion belongs to an admin/moderator use case.

1.2. Use Case Diagram



1.3. Fully Dressed Use Cases

UC-1: Manage Profile and Preferences

Primary Actor: End User

Goal in Context: The user wants to update personal profile information, selected interests, and readability preferences so that the system can generate a more personalized and accessible news experience.

Stakeholders and Interests:

- 1) End User: Wants to manage personal information, choose interests, and adjust interface settings such as dark mode or font size.
- 2) System: Wants to store accurate user profile and preference data for personalization.
- 3) Recommendation System: Needs updated interest data to generate better personalized news feeds.

Preconditions:

- 1) The user is registered in the system.
- 2) The user is logged in securely.
- 3) The profile and preference management page is available.

Success Guarantee: The user's profile information, selected interests, dark mode setting, and font size preference are saved successfully. The updated preferences are used in future personalized feed generation.

Main Success Scenario:

- 1) The user opens the Profile and Preferences page.
- 2) The system displays the current profile information and saved preferences.
- 3) The user updates profile information such as name or account details.
- 4) The user selects or updates news interest categories.
- 5) The user changes readability settings such as dark mode or adjustable font size.
- 6) The system validates the entered information.
- 7) The system saves the updated profile information and preferences.
- 8) The system confirms that the changes were saved successfully.
- 9) The updated preferences are used for future personalized news feed generation.

Extensions:

2a. Profile information cannot be loaded

- 1) The system fails to retrieve the current profile information.
- 2) The system displays an error message.
- 3) The user is asked to refresh the page or try again later.

6a. Invalid or incomplete profile information

- 1) The user enters missing or invalid information.
- 2) The system highlights the incorrect fields.
- 3) The user corrects the information.
- 4) The system validates the information again.

7a. Preference update fails

- 1) The system cannot save the updated preferences because of a database or server error.
- 2) The system keeps the previous preferences unchanged.
- 3) The system informs the user that the update failed.

UC-2: View Personalized Feed

Primary Actor: End User

Goal in Context: The user wants to view a personalized digital newspaper feed generated according to selected interests, previous interactions, trending topics, relevance, and publication metadata.

Stakeholders and Interests:

- 1) End User: Wants to see relevant, updated, and personalized news in a digital newspaper format.
- 2) System: Wants to generate a ranked and organized personalized feed.
- 3) Recommendation System: Wants to use reading behavior and previous interactions to improve personalization.
- 4) Administrator: Wants engagement statistics to be generated from user activity.

Preconditions:

- 1) The user is logged in securely.
- 2) The user has selected interests or profile preferences.
- 3) News articles have already been collected, classified, and stored.
- 4) Ranking or recommendation data is available, or the system can use a fallback ranking method.

Success Guarantee: The user views a personalized digital newspaper feed with relevant articles, prioritized headlines, source references, publication dates, trending news, recommendations, and integrated institutional events where applicable.

Main Success Scenario:

- 1) The user opens the Personalized Feed page.
- 2) The system retrieves the user's selected interests and profile preferences.
- 3) The system retrieves previous user interactions and reading behavior.
- 4) The system selects candidate news articles according to the user's interests.
- 5) The system recommends articles based on previous interactions.
- 6) The system analyzes trending topics and highlights trending news.
- 7) The system adapts the content ranking according to user behavior and preferences.
- 8) The system prioritizes headlines automatically based on relevance or importance.
- 9) The system generates a digital newspaper layout for the personalized content.
- 10) The system includes source references and publication dates for the articles.
- 11) The system integrates important institutional events into the personalized newspaper edition when applicable.
- 12) The system displays the daily or weekly newspaper edition.
- 13) The system records engagement statistics related to user activity.

Extensions:

4a. No articles match the user's interests

- 1) The system cannot find enough articles related to the selected interests.
- 2) The system displays a message explaining that limited content is available.
- 3) The system may show general trending news or ask the user to update preferences.

7a. Recommendation data is unavailable

- 1) The system cannot access previous interaction or ranking data.
- 2) The system uses selected interests, recency, or popularity as a fallback.
- 3) The personalized feed is still displayed.

11a. No important institutional events are available

- 1) The system checks for institutional events.
- 2) No relevant events are found.
- 3) The system displays the personalized newspaper without event content.

UC-3: Search and Filter News

Primary Actor: End User

Goal in Context: The user wants to search news articles by keyword, apply advanced filters, view trending news, bookmark/save articles, and mark articles as read or unread.

Stakeholders and Interests:

- 1) End User: Wants to quickly find, filter, save, and manage articles.
- 2) System: Wants to provide accurate search results and update article interaction data.
- 3) Recommendation System: Wants to track user reading behavior for personalization.
- 4) Analytics System: Wants to use search and interaction data to improve future recommendations.

Preconditions:

- 1) The user is logged in securely.
- 2) The system contains searchable news articles.
- 3) The search and filtering functions are available.

Success Guarantee: The user receives relevant search results, applies filters successfully, saves/bookmarks articles when desired, and updates article read/unread status.

Main Success Scenario:

- 1) The user opens the Search and Filter News page.
- 2) The user enters a keyword into the search bar.
- 3) The system performs keyword-based search.
- 4) The system displays matching news articles.
- 5) The user applies advanced filters such as category, date, source, or popularity.
- 6) The system updates the result list according to the selected filters.
- 7) The system detects and highlights trending news in the results where applicable.
- 8) The user opens a selected article.
- 9) The system tracks the user's reading behavior for personalization purposes.
- 10) The user bookmarks or saves the article.
- 11) The user marks the article as read or unread.
- 12) The system saves the user's article interaction status.

Extensions:

3a. No search results are found

- 1) The system cannot find articles matching the keyword.
- 2) The system displays a no-results message.
- 3) The system suggests changing the keyword or removing filters.

5a. Filters return an empty result list

- 1) The user applies too many restrictive filters.
- 2) The system shows that no articles match the selected filters.
- 3) The user clears or changes the filters.
- 4) The system refreshes the result list.

10a. Bookmark saving fails

- 1) The user tries to save an article.
- 2) The system cannot save the bookmark because of a server or database problem.
- 3) The system displays an error message and does not mark the article as saved.

UC-4: Export and Print Hardcopy

Primary Actor: End User

Goal in Context: The user wants to export the personalized newspaper as a print-ready PDF, preview it before printing, choose a layout template, and download or print it directly.

Stakeholders and Interests:

- 1) End User: Wants to obtain a printable version of the personalized newspaper.
- 2) System: Wants to generate a correct print-ready PDF file.

Preconditions:

- 1) The user is logged in securely.
- 2) A personalized newspaper edition or feed is available.
- 3) The export and print service is available.

Success Guarantee: The system generates a print-ready PDF file in the selected layout format. The user can preview, download, or directly print the personalized newspaper.

Main Success Scenario:

- 1) The user opens the personalized newspaper edition.
- 2) The user selects the Export / Print option.
- 3) The system displays available layout templates such as A4, tabloid, and booklet formats.
- 4) The user selects a layout template.
- 5) The user requests a print preview.
- 6) The system generates and displays the print preview.
- 7) The user confirms the export.
- 8) The system generates the personalized newspaper in print-ready PDF format.
- 9) The system provides download and direct print options.
- 10) The user downloads the PDF or sends it directly to the printer.

Extensions:

1a. No personalized newspaper edition is available

- 1) The user tries to export a newspaper.
- 2) The system detects that there is no available newspaper edition.
- 3) The system informs the user and redirects the user to the personalized feed page.

6a. Print preview cannot be generated

- 1) The system fails to create the print preview.
- 2) The system displays an error message.
- 3) The user may retry or choose another layout template.

8a. PDF generation fails

- 1) The system cannot generate the PDF file.
- 2) The system cancels the export operation.
- 3) The system informs the user and offers a retry option.

UC-5: View Institutional Events and Announcements

Primary Actor: End User

Goal in Context: The user wants to view important institutional events and announcements, receive reminders or notifications, and see relevant events integrated into the personalized newspaper edition.

Stakeholders and Interests:

- 1) **End User:** Wants to stay informed about academic, social, deadline, and exam-related events.
- 2) **System:** Wants to display, categorize, notify, remind, and integrate events into personalized newspaper editions.
- 3) **Notification Service:** Delivers push or email notifications for critical announcements.

Preconditions:

- 1) The user is logged in securely.
- 2) Institutional events or announcements have been detected or published.
- 3) The event and announcement module is available.

Success Guarantee: The user views important institutional events and announcements, sees their categories, receives reminders or notifications where applicable, and relevant events are integrated into the personalized newspaper edition.

Main Success Scenario

- 1) The user opens the Events and Announcements section.
- 2) The system automatically detects and displays important institutional events.
- 3) The system categorizes events as academic, social, deadlines, or exams.
- 4) The user views the list of events and announcements.
- 5) The user selects an event to see detailed information.
- 6) The system displays event details such as title, category, date, and description.
- 7) The system sends push or email notifications for critical announcements when needed.
- 8) The system provides event reminder functionality.
- 9) The system integrates important events into the personalized newspaper edition.
- 10) The user returns to the event list or opens the personalized newspaper edition.

Extensions:

2a. No institutional events are available

- 1) The system checks for important institutional events.
- 2) No event or announcement is found.
- 3) The system displays an empty-state message.

7a. Notification delivery fails

- 1) The system tries to send a push or email notification.
- 2) The notification service fails or becomes unavailable.
- 3) The system keeps the event visible in the application.
- 4) The system may retry notification delivery later.

8a. User does not want reminders

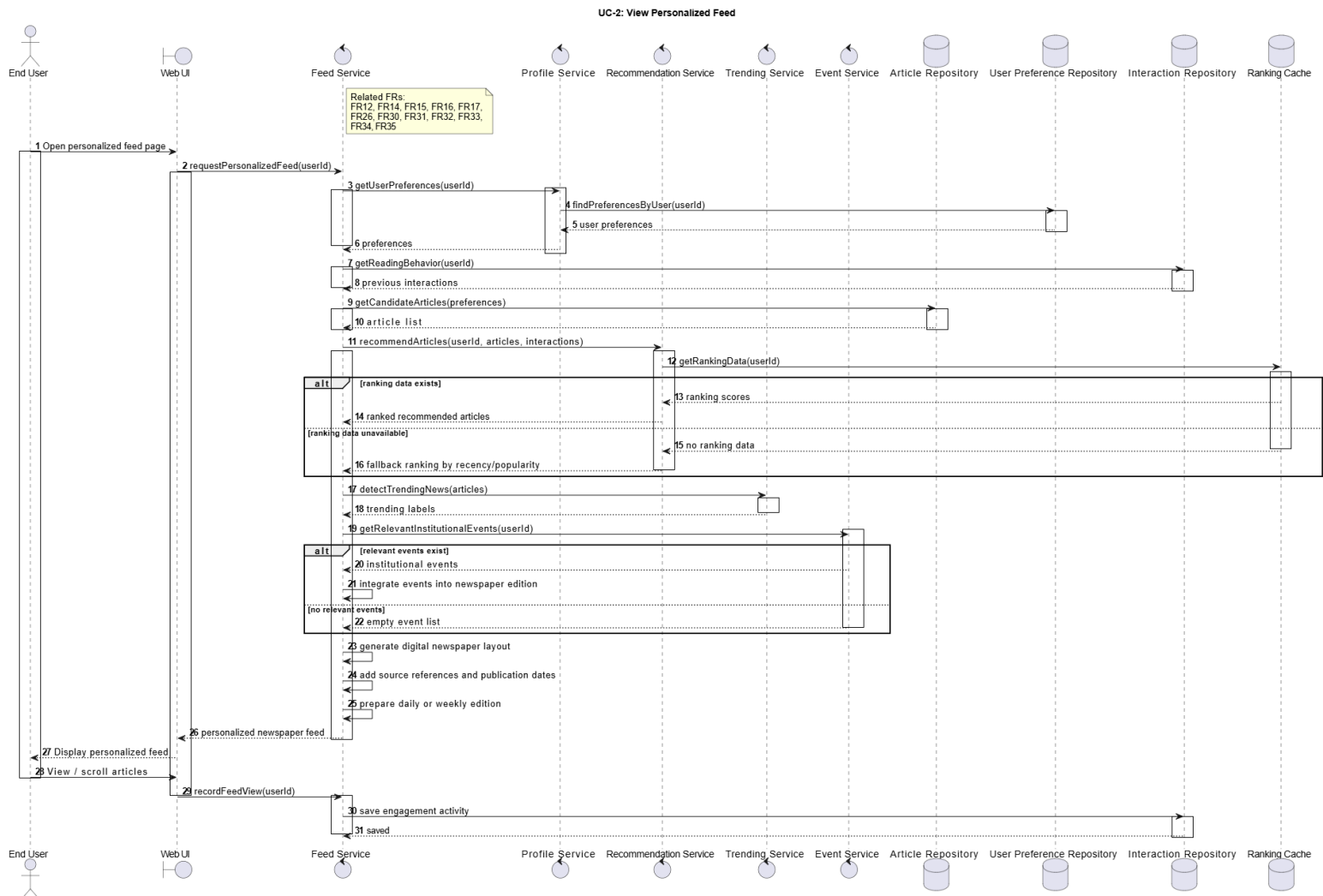
- 1) The system offers reminder functionality.
- 2) The user disables or ignores reminders.
- 3) The event remains visible, but no reminder is activated for that user.

2. Sequence Diagram

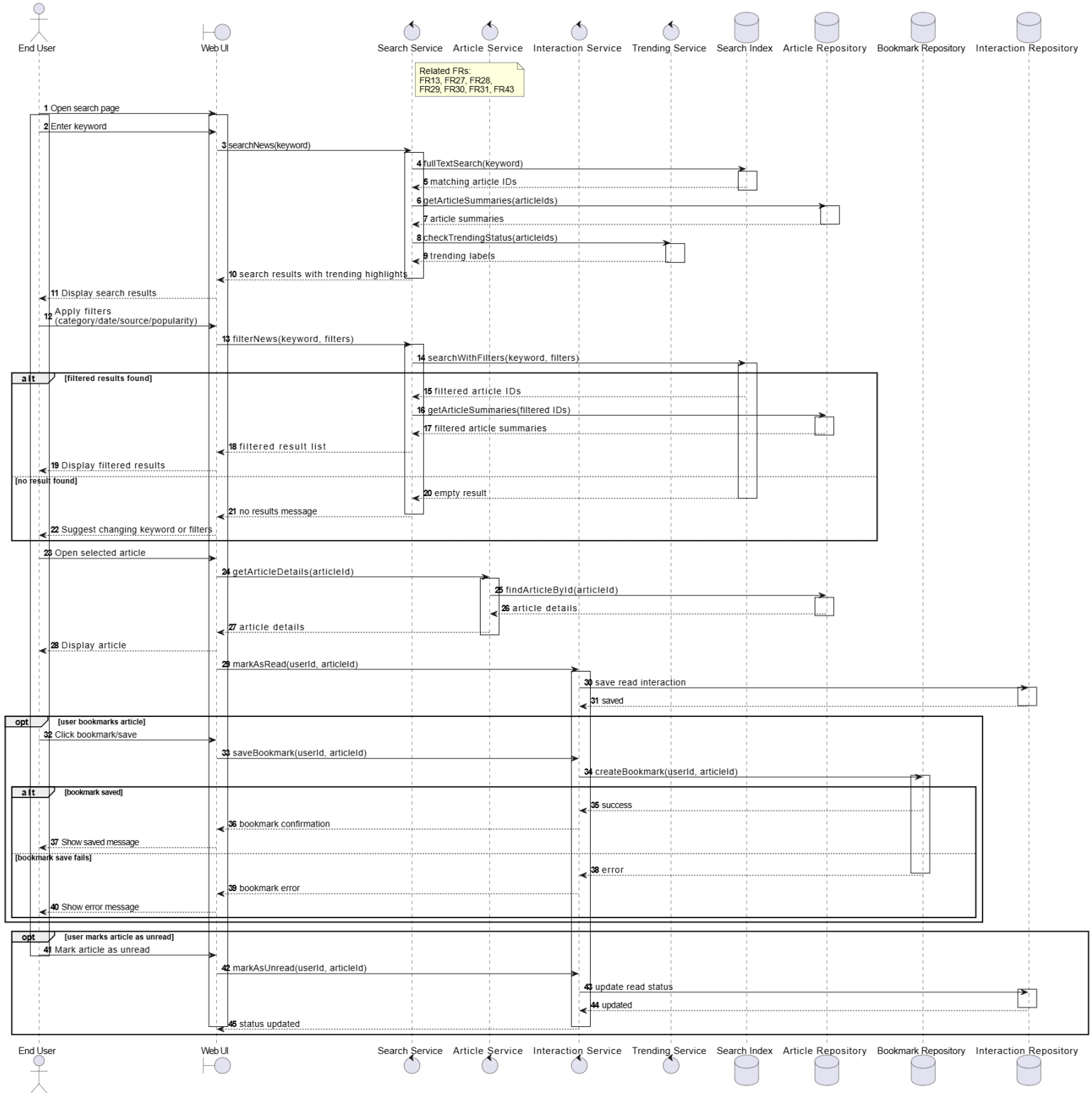
2.1. Use Case Selection

UC2: View Personalized Feed

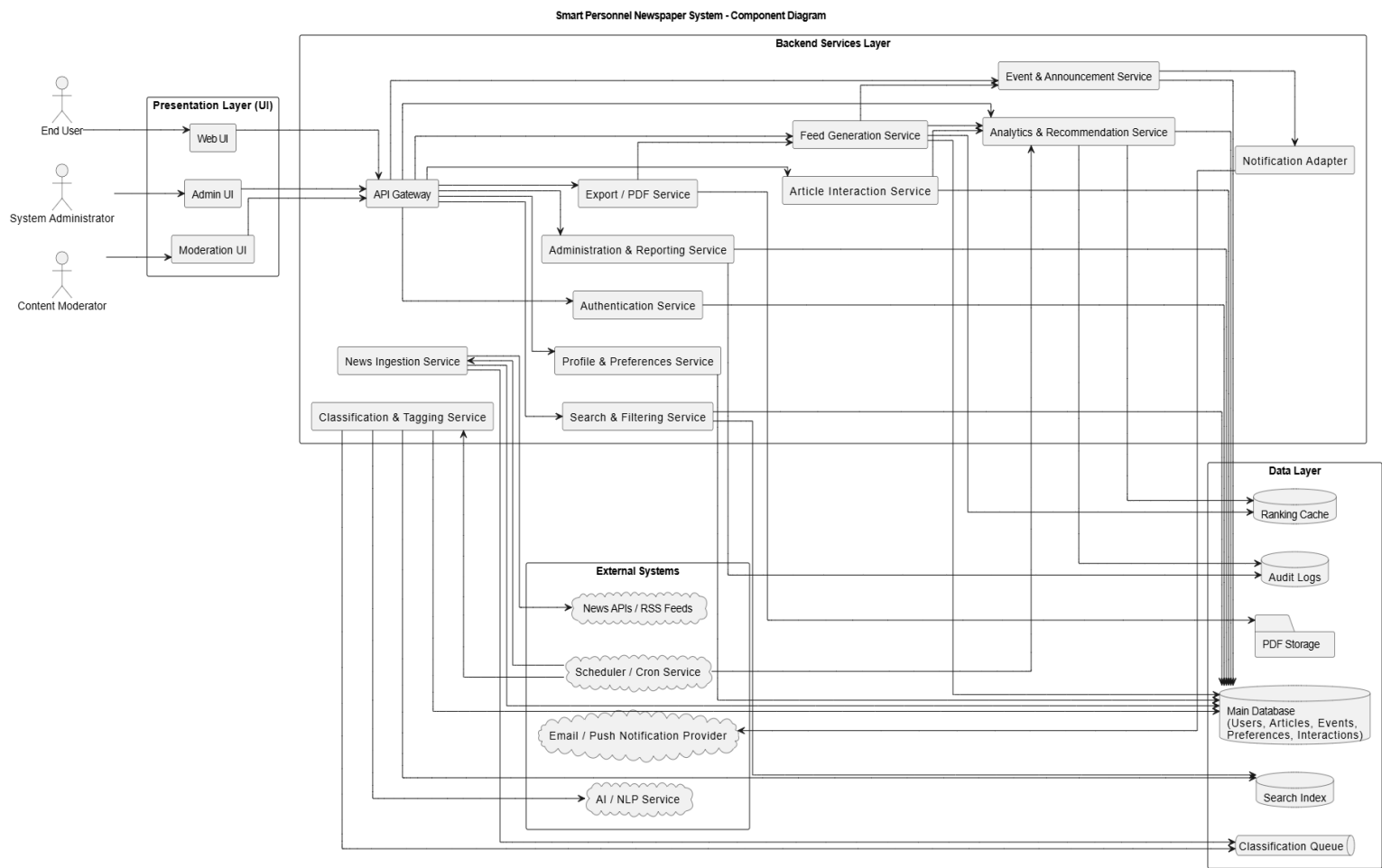
UC3: Search and Filter News



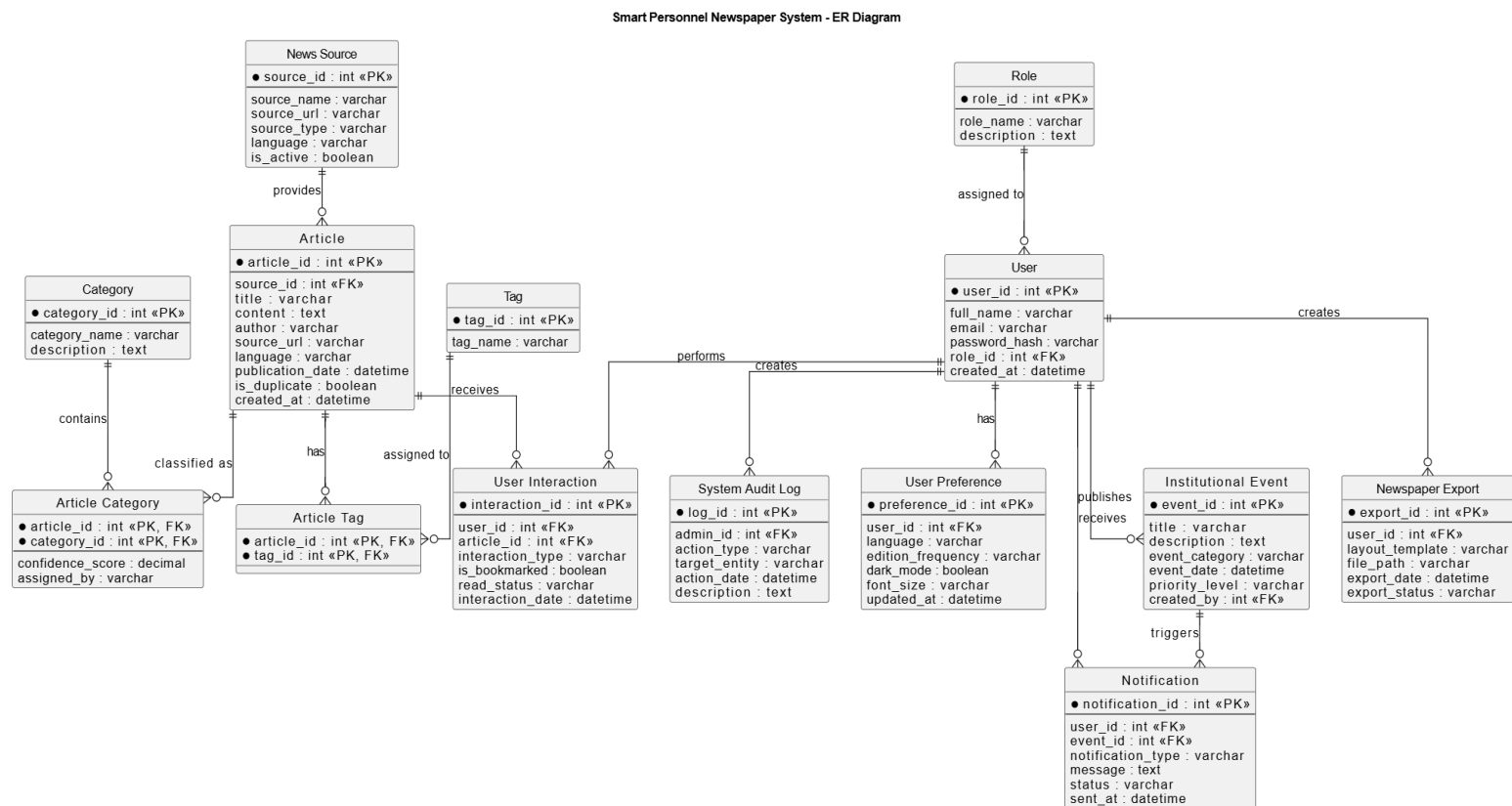
UC-3: Search and Filter News



3. Component Diagram



4. Data Diagram



5. System Architecture

5.1. Architecture Description

The Smart Personnel Newspaper system follows a layered client-server architecture. This architecture separates the system into different layers such as the presentation layer, backend services layer, data layer, and external integration layer. This separation makes the system easier to develop, maintain, test, and extend.

In the client-server architecture, users interact with the system through the web-based user interface. The client side includes the Web UI, Admin UI, and Moderation UI. These interfaces send requests to the server side through an API Gateway. The backend services process user requests, apply business logic, communicate with databases, and return the required results to the client.

The layered architecture is used to organize the system into clear responsibilities. The Presentation Layer handles user interaction. The Backend Services Layer includes authentication, profile management, personalized feed generation, search and filtering, article interaction, PDF export, event management, analytics, recommendation, news ingestion, classification, and administration services. The Data Layer stores users, articles, preferences, interactions, events, notifications, export records, search indexes, ranking data, and audit logs. The system also uses external integration components because it depends on outside services. News APIs and RSS feeds are used for collecting articles. AI/NLP services are used for classification, tagging, and recommendation support. Email or push notification providers are used for sending announcements and reminders. A scheduler or cron service is used to trigger automated background tasks such as news fetching, classification, and ranking updates.

This architectural style is suitable for the Smart Personnel Newspaper system because the project includes multiple independent modules such as news collection, classification, personalized newspaper generation, hardcopy printing, event announcements, search and filtering, analytics, recommendation, and administration.